INCH-POUND

MIL-B-48343A (AR)
11 March 1993
SUPERSEDING
MIL-B-48343 (AR)
17 May 1978

MILITARY SPECIFICATION

Battery, Primary Reserve, Ammonia, 5 Milliampere Hour Capacity

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, manufacture and assembly of the battery used in the housing, timing, and fuzing for the projectile, 155MM, M692/M731. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

MILITARY

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specifications For Assemblies.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1345

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STANDARDS

MILITARY

MIL-STD-331	-Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1169	-Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.6)

U.S. ARMY ARMAMENTS RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9298625	-Battery, Assembly

(Copies of drawings, publications and other Government documents required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E498-73	-Methods of Testing for Leaks Using the Mass Spectrometer Leak Detector or Residual Gas Analyzer in the Tracer Probe Mode

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specification.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on dwg. 9298625 and associated drawings and with all requirements specified in applicable specifications.

3.3 Ampule.

3.3.1 Seal. The ampule shall not show evidence of leakage when tested in accordance with 4.5.1.1.

3.3.2 Environmental. The ampule shall be capable of withstanding exposure to the environments listed herein.

3.3.2.1 Temperature cycling. The ampule shall be capable of withstanding temperature cycling from minus sixty-five degrees Fahrenheit plus or minus five degrees Fahrenheit to plus one hundred and seventy degrees Fahrenheit plus five degrees Fahrenheit ($-65^{o}F \pm 5^{o}F$ to $+ 170^{o}F \pm 5^{o}F$) when tested as specified in 4.5.1.2.

3.3.2.2 Time and temperature to fragmentation. The ampule, after an initial stabilization at $160^{o}F$ for at least one hour, shall be exposed to an increasing temperature of $5^{o}F$ (at a rate not greater than $5^{o}F$ per 15 minute period). The ampule shall not be fragmentized at any temperature of $170^{o}F$ or below.

3.3.3 Ammonia composition.

a. The ammonia used to fill the ampule shall be anhydrous and of 99.98% minimum purity.

b. The filled and sealed ampule shall not contain more than .10% water.

3.3.4 Ammonia weight. The weight of ammonia in a sealed ampule shall be seventy plus twenty five (70 + 25) milligrams.

3.4 Cathode.

3.4.1 Moisture content. the cathode material shall not average more than one percent (1.0%) of water. No sample shall have a moisture content greater than one and five tenths percent (1.5%).

3.4.2 Weight. The cathode weight shall be seventy nine plus or minus five (79 ± 5.0) milligrams.

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3.5 Battery primary reserve.

3.5.1 Case seal and pre-initiation voltage. The battery shall be sealed to prevent leakage. Actual leak rate shall not exceed 5x10 -7 cc/Sec. of helium. Reference ASTM E498-73, Method A (The volume for calculating the actual leak rate shall be 0.5 cc.)

3.5.2 Activation. The battery shall be initiated and activated in accordance with 4.5.9.2.

3.5.3 Rise time. The battery output voltage shall rise to three and three tenths plus one and nine tenths (3.3 + 1.9) volts within ten (10) seconds when tested in accordance with 4.5.9.3. During the rise time, the output voltage shall not regress below two and two tenths (2.2) volts after attaining two and seven tenths (2.7) volts.

3.5.4 Short time capacity. The battery shall be activated to the requirements of 3.5.2, meet the rise time requirements of 3.5.3 and have an output voltage of less than two and seventh tenths (2.7) volts after four (4) hours of continuous operation when tested as specified in 4.5.9.4.

3.5.5 Long time capacity. The battery shall be activated to the requirements of 3.5.2, meet the rise time requirements of 3.5.3 and have an output voltage of three and three tenths plus one and nine tenths (3.3 + 1.9) volts for thirty eight (38) (Section I) to forty eight (48) (Section II) hours when tested as specified in 4.5.9.5.

3.5.6 Test temperature. The battery shall be capable of meeting all specified requirements at any temperature from minus thirty two degrees centigrade plus six degrees centrigrade to plus sixty degrees centigrade minus six degrees centigrade (-32°C + 6°C to + 60°C -6°C).

3.6 Environmental and mechanical test.

3.6.1 Thermal shock. The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.1.

3.6.2 Temperature cycling. The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.2.

3.6.3 Transportation vibration. The battery shall be capable of withstanding mechanical shocks when tested in accordance with 4.5.10.3.

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3.6.4 Drop test. The battery shall be capable of withstanding shock when tested as specified in 4.5.10.4.

3.6.5 Mechanical shock. the battery shall be capable of withstanding shock when tested as specified in 4.5.10.5.

3.7 Repair. Any module failing to comply with all the test requirements and undergoes repair shall be tested as specified in 4.5.11.

3.8 Test temperature. Unless specified otherwise, all testing shall be performed at a temperature of twenty five degrees centigrade plus or minus five degrees centigrade. ($25^{o}C \pm 5^{o}C$)

3.9 Data requirements. Data shall be recorded and retained for all test required by this specification and as specified in 4.5.12.

3.10 Process controls. All process controls essential for battery manufacturer as deemed necessary by the Government to assure a reliable end item shall be submitted and maintained in accordance with 4.5.13.

3.11 Workmanship. All mechanical assemblies shall be fabricated and finished in a thorough, workmanlike manner. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agents. All markings and stampings shall be free of harmful cleaning and soldering agents that may be injurious to the assembly. All markings and stampings, either component or assembly shall be displayed and visibly defined. The battery shall be manufactured and processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

3.11.1 Workmanship. The Ampule and Case Assembly shall not have any dented, bulging, or damaged cases.

3.11.2 Workmanship. The Case shall not be crushed or damaged.

3.11.3 Workmanship. The Cover and Collar Assembly shall have all parts present and be free from weld burrs under the paper. The separator paper shall not be torn and the collars shall not be damaged.

3.11.4 Workmanship. The Anodes shall not have sheared edges or surface discoloration.

3.11.5 Workmanship. The Collar and the Cell Cup shall not be damaged or discolored.

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3.11.6 Workmanship. The Cover Assembly shall not be oily, dirty or rusty.

3.11.7 Workmanship. The Cup and Anode Assembly shall be assembled correctly and shall not have any deformation of the cell cup.

3.11.8 Workmanship. The Battery Assembly shall not have any dented, bulging, or cracked cases. It shall not have any pronounced weld burrs or discolored covers.

3.11.9 Workmanship. The Support, Ampule shall not have any cracks or excessive flash.

3.12 Qualification. The battery shall be requalified if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the battery are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocates his assembly facilities.

c. The contractor uses a new supplier for major subassemblies.

d. A change in material used for battery production or assembly.

Qualification requirements shall be determined by the Government for each change as specified in 4.4.3.14.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3)

b. Quality conformance inspection. (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below:

PART DESCRIPTION	DRAWING NO.	QUANTITY
Ampule and Case Assembly	9298626	25
Bulkhead	9298627	25
Ampule, Glass	9298628	75
Case	9298629	25
Separator	9298631-1	25
Separator	9298631-2	25
Cover and Collar Assembly	9298632	25
Anode No. 1	9298633-1	25
Anode No. 2	9298633-2	25
Collar	9298634	25
Cover Assembly	9298636	25
Cathode	9298640	75
Cup and Anode Assembly	9298642	25
Cell Cup	9298645	25
Battery Primary Reserve, Assembly	9298625	336/672
Support, Ampule	9298650	25

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4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein and the following paragraph.

4.3.2.1 Battery assembly test. Battery assemblies listed in Table I shall be tested as decribed below (See Fig. 1):

a. Thirty six (36) batteries shall be subjected to the five (5) environmental tests in the sequence shown. After the environmental testing, the batteries shall be subjected to the electrical test which includes activation (4.5.9.2), rise time (4.5.9.3) and long time capacity (4.5.9.5). However, the batteries will not require rotation during activation. Twelve (12) each batteries shall be tested at the temperature listed in Table II of 4.4.3.7.4. If the test sample contains one(1) reject for activation, rise time or Section I of the long time discharge requirements, a second sample of 36 batteries shall be tested. The second sample shall have passed the above criteria providing there are zero (0) rejects in the sample. The test shall be rejected if the first sample or the combined sample contain two (2) or more rejects for activation, rise time or discharge requirements of 4.4.3.7.2.

b. Two hundred and twenty five (225) shall be subjected to activation, rise time and long time capacity. Seventy five (75) each batteries shall be tested at the three temperatures listed in Table III of 4.4.3.7.4. one (1) reject for the first sample or three (3) rejects for the combined sample shall be allowed for activation and rise time requirement or Section I discharge requirements of 4.4.3.7.2. Three (3) rejects for the first sample or four (4) rejects for the combined sample shall be allowed through Section II discharge requirement of 4.4.3.7.3.

c. Twenty five (25) each batteries shall be tested for short time capacity at the three temperatures listed in the Table III of 4.4.3.7.4. One (1) reject shall be allowed for the requirement of 4.4.3.7 and one (1) when tested as specified in 4.5.9.4.

4.3.3 Rejection. See MIL-A-48078. The First Article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or test specified in this specification or any of the criteria specified in 4.3.2.1 above.

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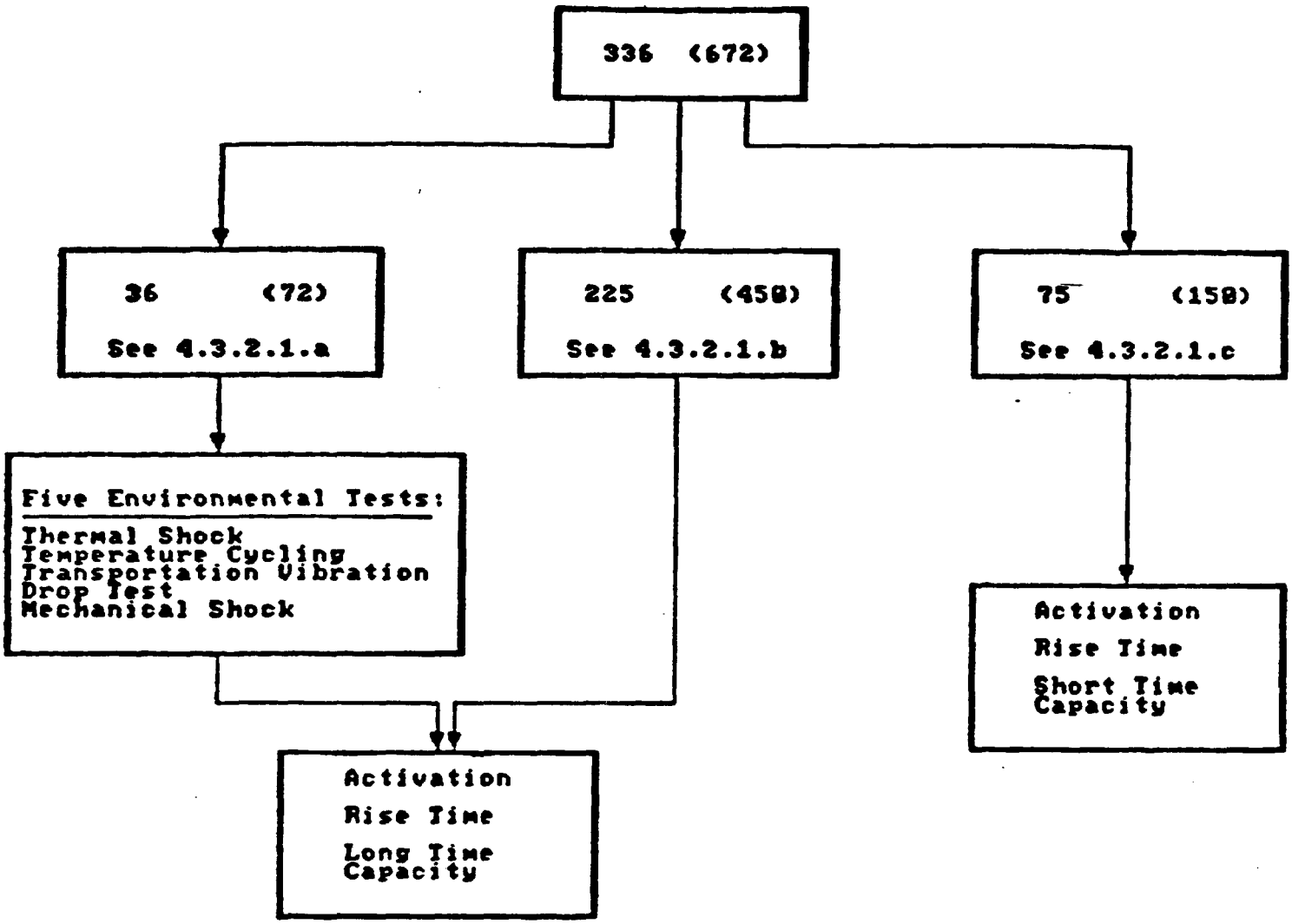


FIGURE 1. First Article Inspection Test Sequence for Battery Assembly.

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 1 OF 4	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Ampule and Case Assembly (Dwg. No. 9298626) Presence of ammonia in ampule	25	3.2	4.5.8
	Bulk head to case force test	25	3.2	4.5.3
	Bulkhead (Dwg. No. 9298627) Examination for defects	25	3.2	4.4.2.2
	Ampule, Glass (Dwg. No. 9298628) Examination for defects	25	3.2	4.4.2.3
	Seal	25	3.3.1	4.5.1.1
	Temperature cycling (short term)	25	3.3.2.1	4.5.1.2.3
	Ammonia composition	25	3.2	4.5.1.3
	Weight	25	3.2	4.5.1.4
	Temperature cycling (long term)	100	3.3.2.1	4.5.1.2.1
	Time & Temperature to fragmentation	25	3.3.2.2	4.5.1.2.2
	Case (Dwg. 9298629) Examination for defects	25	3.2	4.4.2.4

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 2 OF 4	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / INSPECTION METHOD REFERENCE
	Separator (Dwg. 9298631-1)		
	Salt content	25	3.2 / 4.5.7
	Separator (Dwg. 9298631-2)		
	Salt content	25	3.2 / 4.5.7
	Cover and collar assembly (Dwg. 9298632)		
	Weld strength	25	3.2 / 4.5.6.1
	Vacuum pull test	25	3.2 / 4.5.6.2
	Anode No. 1 (Dwg. 9298633-1)		
	Examination for defects	25	3.2 / 4.5.2.6
	Anode No. 2 (Dwg. 9298633-2)		
	Examination for defects	25	3.2 / 4.4.2.7
	Collar (Dwg. 9298634)		
	Examination for defects	25	3.2 / 4.4.2.8

NOTES:

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 3 OF 4	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cover assembly (Dwg. 9298636) Examination for defects	25	3.2	4.4.2.9
	Cathode (Dwg. 9298640) Moisture content Weight Composition	25 25 25	3.4.1 3.4.2 3.2	4.5.2.1 4.5.2.2 4.5.2.3
	Cup and Anode Assembly (Dwg. 9298642) Weld test Vacuum pull test	25 25	3.2 3.2	4.5.5.1 4.5.5.2
	Cell Cup (Dwg. 9298645) Examination for defects	25	3.2	4.4.2.11
	Support Ampul (Dwg. 9298650) Examination for defects Hardness test	25 Note a, 336	3.2	4.4.2.13

NOTES:

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 4 OF 4	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Battery assembly (Dwg. 9298625)	Note b,		
	Thermal shock	36	3.6.1	4.5.10.1/4.3.2.a
	Temperature cycling	36	3.6.2	4.5.10.2/4.3.2.a
	Transportation vibration	36	3.6.3	4.5.10.3/4.3.2.a
	Drop test	36	3.6.4	4.5.10.4/4.3.2.a
	Mechanical shock	36	3.6.5	4.5.10.5/4.3.2.a
	Activation	225	3.5.2	4.5.9.2/4.3.2.b
	Rise time	225	3.5.3	4.5.9.3/4.3.2.b
	Long-time capacity	225	3.5.5	4.5.9.5/4.3.2.b
	Short-time capacity	75	3.5.4	4.5.9.4/4.3.2.c
	Case seal and pre-initiation voltage	25	3.5.1	4.5.9.1

NOTES:
a. A coupon from the support ampule batch shall be used as the sample.
b. Double sample may apply. See 4.3.2

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Where cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE I Attributes sampling inspection.

Lot Size	Inspection Level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.4). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ampule and Case Assemlby	1 OF 1	9298626
			NEXT HIGHER ASSEMBLY 9298647

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of ammonia in ampule	100%	3.2	4.5.8
102	Bulkhead to case force test		3.2	4.4.3.9/4.5.3
Minor				
201	Poor workmanship	Level V	3.11.1	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bulkhead	1 OF 1	9298627
			NEXT HIGHER ASSEMBLY: 9298646

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Thickness from outside to bottom of spherical radius	Level III	3.2	Gage
Minor 201	Poor workmanship	Level V	3.11	Visual

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Ampule, Glass	1 OF 1	9298628
			NEXT HIGHER ASSEMBLY 9298626

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Seal	Level III	3.2.1	4.5.1.1
102	Temperature cycling	100%	3.2.2	4.5.1.2.3
103	Ammonia composition		3.2.3	4.4.3.8/4.5.1.3
104	Depth from base to mid-point at top of shoulder	Level III	3.2	Gage
105	Depth of center point	Level III	3.2	Gage
106	Concentricity of tip	Level III	3.2	Gage
107	Presence of ammonia in ampule	100%	3.2	Visual
108	Ammonia weight	Level III	3.2	4.5.1.4
Minor				
201	Poor workmanship	Level V	3.11	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Case	1 OF 1	9298629
			NEXT HIGHER ASSEMBLY 9296626

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside radius at bottom	Level III	3.2	Gage
102	Thickness of bottom	Level III	3.2	Gage
Minor				
201	Flatness of flange	Level V	3.2	Gage
202	Poor workmanship	Level V	3.11.2	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cover and Collar Assembly	1 OF 1	9298632
			NEXT HIGHER ASSEMBLY 9298647

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Vacuum pull test	100%	3.2	4.5.6.2
102	Weld strength		3.2	4.4.3.4/4.5.6.1
Minor				
201	Poor workmanship	Level V	3.11.3	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Anode No. 1	1 OF 1	9298633-1
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter	Level V	3.2	Gage
202	Poor workmanship	Level V	3.11.4	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Anode No. 2	1 OF 1	9298633-2
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter	Level V	3.2	Gage
202	Poor workmanship	Level V	3.11.4	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-48343A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Collar	1 OF 1	9298634
			NEXT HIGHER ASSEMBLY 9298632

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Web thickness	Level III	3.2	Gage
Minor				
201	Anode side inside diameter, min.	Level V	3.2	Gage
202	Cover assembly inside diameter. max.	Level V	3.2	Gage
203	Poor workmanship	Level V	3.11.5	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Cover Assembly	1 OF 1	9298636
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Depth of projection diameter	Level III	3.2	Gage
Minor 201 203	Diameter of small outside diameter Poor workmanship	Level V Level V	3.2 3.11.6	Gage Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Cup and Anode Assembly	1 OF 1	9298642
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Vacuum pull test	100%	3.2	4.4.5.2
102	Weld strength		3.2	4.4.3.3/4.5.5.1
Minor				
201	Poor workmanship	Level V	3.11.7	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Cell Cup	1 OF 1	9298645
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total Height	Level III	3.2	Gage
Minor				
201	Presence of slots and holes	Level V	3.2	Visual
202	Poor workmanship	Level V	3.11.5	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Battery Assembly	1 OF 1	9298625
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of case	Level III	3.2	Gage
102	Overall height	Level III	3.2	Gage
103	Case seal (pre-initiation voltage)	100%	3.5.1	4.5.9.1
104	Activation		3.5.2	4.5.9.2
105	Rise time		3.5.3	4.4.3.7.1/4.5.9.3
106	Long term capacity Section I		3.5.5	4.4.3.7.2/4.5.9.5
107	Long term capacity Section II		3.5.5	4.4.3.7.3/4.5.9.5
Minor				
201	Poor workmanship	Level V	3.11.8	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Support, Ampule	1 OF 1	9298650
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hardness Test		3.2	4.4.3.2/4.5.4
Minor				
201	Inside diameter, minimum	Level V	3.2	Visual
202	Thickness	Level V	3.2	Gage
203	Poor workmanship	Level V	3.11.9	Visual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Cathode (See 9298640).

4.4.3.1.1 Moisture content. Major Defect - Each batch of cathode material shall be sampled for the specified moisture requirements in accordance with 4.5.2.1. If the batch passes the test requirements, it may be stored for a maximum of ninety (90) days providing the maximum relative humidity at ambient does not exceed five percent (5.0%). The batch shall be rejected if the sample is classed defective.

4.4.3.1.2 Weight, Major defect. Five (5) cathodes from each station shall be weighed each production shift in accordance with 4.5.2.2. The previously made cathodes shall be rejected if one (1) or more samples are classed defective.

4.4.3.1.3 Composition, Major defect. Each batch shall be tested for the specified requirement in accordance with 4.5.2.3. If the batch passes the test requirements, it may be stored for a maximum of ninety (90) days providing the maximum relative humidity does not exceed five percent (5%). The batch shall be rejected if the sample is classed defective.

4.4.3.2 Support ampule hardness (See 9298650). Major defect. - A coupon specimen from each support ampule batch shall be made and tested in accordance with 4.5.4. The lot shall be rejected if the coupon is rejected.

4.4.3.3 Cup and anode assembly weld strength. (See 9298642) Major defect. - Each welder shall be monitored for control and deviation in accordance with the following procedures and as specified in 4.5.5.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after forty eight (48) consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

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The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in eighty (80) consecutive hours of production. If one (1), or more samples are classed defective, the prior two (2) hours production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.4 Cover and collar assembly weld strength. (See 9298632) Major defect. - Each welder shall be monitored for control and deviation in accordance with the following procedures and as specified in 4.5.6.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after forty eight (48) consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in eighty (80) consecutive hours of production. If one (1) or more samples are classed defective the prior two (2) hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.5 Separator No. 1 salt content, (See 9298331-1), Major defect. - Each batch of separators shall be tested for the specified requirements in accordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

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4.4.3.6 Separator No. 2 salt content. (See Dwg. 9298331-2), Major Defect. Each batch of separators shall be tested for the specified requirements in acordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

4.4.3.7 Battery, reserve, primary (See dwg. 9298647).

4.4.3.7.1 Activation and rise time. Major defect. One hundred and twenty-five (125) batteries shall be randomly selected from each lot containing less than ten thousand (10,000) units. The batteries shall be tested for activation in accordance with 4.5.9.2 and rise time in accordance with 4.5.9.3. The lot shall be accepted if one (1) or less defects are in the above sample.

If the test sample contains two (2) rejects a second sample of one hundred and twenty-five (125) batteries shall be randomly selected from the lot. The units shall be tested as above for the activation and rise time requirements.

The lot shall be rejected if the first sample contains three (3) or more defects or if the combined total of both samples yields four (4) or more defects.

Test temperatures and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

One hundred and seventy-five (175) batteries shall be randomly selected from each lot containing ten thousand (10,000) or more units. The samples shall be tested for activation in accordance with 4.5.9.2 and rise time in accordance with 4.5.9.3. The lot shall be accepted if one (1) or less defects are in the above sample. If the test sample contains two (2) rejects, a second sample of one hundred and seventy-five (175) batteries shall be randomly selected from the lot. These units shall be tested as specified above for the activation and rise time requirements. The lot shall be rejected if the first sample contains three (3) or more defects or if the combined total of both samples yields five (5) or more defects.

Test temperatures and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.2 Long time capacity discharge section I. Major Defect. for lot sizes less than ten thousand (10,000) units, the contractor shall submit a sample of one hundred and twenty-five (125) batteries which have successfully passed the requirements of 4.4.3.7.1 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if one (1) or less samples are classed defective.

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If the test sample contains two (2) rejects, a second sample of one hundred and twenty-five (125) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

The lot shall be rejected if the first sample contains three (3) or more rejects or if the combined total of both samples yields four (4) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

For lot sizes of ten thousand (10,000) or more batteries, the contractor shall submit a sample of one hundred and seventy-five (175) batteries which have successfully passed the requirements of 4.4.3.7.1 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if one (1) or less samples are classed defective.

If the test sample contains two (2) rejects a second sample of one hundred and seventy-five (175) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

The lot shall be rejected if the first sample contains three (3) or more rejects or if the combined total of both samples yields five (5) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.3 Long time capacity discharge, section II. Major Defect. For lot sizes less than ten thousand (10,000) units, the contractor shall submit a sample of one hundred and twenty-five (125) batteries which have successfully passed the requirements of 4.4.3.7.2 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if four (4) or less samples are classed defective. If the test sample contains five (5) or six (6) rejects, a second sample of one hundred and twenty-five (125) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

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The lot shall be rejected if the first sample contains seven (7) or more rejects or if the combined total of both samples yields ten (10) or more rejects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

For lot sizes of ten thousand (10,000) or more batteries, the contractor shall submit a sample of one hundred and seventy-five (175) batteries which have successfully passed the requirements of 4.4.3.7.2 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if six (6) or less samples are classed defective.

If the lot contains more than six (6) defects but less than thirteen (13) defects, a second sample of one hundred and seventy-five (175) batteries shall be selected from the lot and tested as above for all specified requirements.

The lot shall be rejected if the first sample contains thirteen (13) or more rejects or if the combined total of both samples yields fourteen (14) or more defects.

4.4.3.7.4 Test temperature and temperature groups.

Test temperature and the number of samples per temperature group shall be in accordance with Table III and Table IV.

TABLE III

For sample size of 125 batteries

Test temperature	-32°C to -26°C	20°C to 30°C	54°C to 60°C
Samples/group 1	42	41	42

TABLE IV

For sample size of 175 batteries

Test temperature	-32°C to -26°C	20°C to 30°C	54°C to 60°C
Samples/group	58	58	59

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4.4.3.8 Ampule weight (See dwg. 9298628), Major Defect. Twenty (20) samples shall be randomly selected from each lot and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be accepted if it contains no defects or rejected if it contains two (2) or more defects. In the event the lot contains one (1) defect, a second sample of twenty (20) ampules shall be randomly selected and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be rejected if the second sample contains one (1) or more defects.

4.4.3.9 Bulkhead to case force test, (See dwg. 9298626) Major Defect. The lot shall be tested 100% for the specified requirements in accordance with 4.5.3 when the contractor uses non-automated manufacturing techniques. Any assembly classed defective shall be removed from the lot.

The sampling shall be reduced whenever the contractor uses a fully automated manufacturing technique that does not lend itself to process interruption. For this procedure, the contractor shall select the last six (6) assemblies produced during the previous hour. The samples shall be tested for the specified requirements in accordance with 4.5.3. The quantity produced during the hour represented by the sample size shall be classed defective if one (1) or more of the samples are classed defective.

4.4.3.10 Repair. Major defect. - All batteries undergoing repair shall be processed in accordance with 4.5.11. Any battery classed defective shall not be returned to the lot.

4.4.3.11 Test temperature. Major defect. - All batteries tested shall be conducted at the specified test temperature. Any battery test performed otherwise shall be classed defective.

4.4.3.12 Data requirements. Major defect.- All battery tests shall have the test data recorded in accordance with 4.5.12. Any material tested without the proper data acquisition shall be classed defective.

4.4.3.13 Process controls. Major defect. - All process control procedures shall be in accordance with 4.5.13. Any process control used that is classed defective shall be cause for lot rejection.

4.4.3.14 Qualification requirements. The battery shall be inspected and tested for all the requirements specified in Table I unless specified otherwise by the technical agency listed in 6.5.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

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4.5 Methods of inspection.

4.5.1 Ampule.

4.5.1.1 Seal. The ampule shall be isolated and completely submerged in a test solution of phenolphthalein for a minimum time of four (4) hours. The test solution shall be composed of twenty five cubic centimeters (25cc) of a master solution dissolved in one thousand (1000) cc of deionized water. The referenced master solution shall consist of one (1) gram of phenolphthalein dissolved in sixty (60) cc of ethyl alchohol diluted to one hundred (100) cc with deionized water.

The ampule shall be classed defective if the test solution surrounding the ampule changes to a red or pink color. (Non-destructive test)

4.5.1.2 Environmental.

a. The capability of the ampule(s) to withstand exposure to the environments shall be substantiated by an evaluation of their performance or condition prior to, during, and after exposure to the environments specified in the table of inspections.

b. In addition, to the tests or examinations after the environmental exposure(s) specified in the tables of inspection, the ampules shall be visually examined to assure that they have survived the exposure(s) without physical damage that would impair their form or fit.

4.5.1.2.1 Temperature cycling (long term). The ampules shall be tested as specified in MIL-STD-331, Test 105. The following exceptions and details shall apply:

a. Humidity requirement of MIL-STD-331 not required.

b. Method of Mounting: The ampule(s) may be mounted in any suitable fixture that will provide isolation from one to another and allow appropriate heat transfer to the ampule(s). The isolation shall be sufficient to prevent adjacent ampule(s) from self destructing or damage from shock and fragmentation from other failing ampule(s) in the same test.

c. Measurements after test: See 4.5.1.2b.

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4.5.1.2.2 Time and temperature to fragmentation.

a. Mount the components as specified in 4.5.1.2

b. Control the test temperature as indicated in 3.3.2.2.

c. Determine time and temperature of fragmentation.

d. Calculate results for conformance to 3.3.2.2.

e. Use caution when testing.

4.5.1.2.3 Temperature cycling (short term). The ampule shall be temperature cycled as specified when mounted in an isolated fixture in the temperature chamber. The ampule shall be cycled five (5) times as follows:

a. Heat soaked for a minimum of one and five tenths (1.5) hours at a minimum temperature of plus one hundred and seventy degrees fahrenheit (+170°F).

b. The temperature shall be gradually reduced.

c. The ampule shall be cold soaked for a minimum of one (1) hour at a maximum temperature of minus sixty-five degrees fahrenheit (-65°F).

d. This is the completion on one (1) cycle.

e. The ampule temperature shall be gradually raised and the start of the second cycle and each successive cycle shall commense. Each cycle shall be in accordance with steps a through c. The five cycles shall be completed within seventy-two (72) hours.

Any ampule that cracks, ruptures, discolors or leaks shall be classed defective and removed from the lot. (Non-destructive test)

4.5.1.3 Ammonia composition. The ampule shall be fractured in a closed container and the vapor shall be analyzed for the specified amount of water using a gas chromatographic method or equivalent. Any ampule failing to comply with all of the specified requirements shall be classed defective (Destructive test).

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4.5.1.4 Ammonia weight. The weight shall be determined by weighing the sample ampule and then fracturing it in a closed container. The vapor shall be allowed to vent and the residue weighed. The difference in weight between the original weight and the residue shall be considered the amount of ammonia contained in the submitted sample. Any ampule failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.2 Cathode.

4.5.2.1 Moisture content. The sample of cathode material shall be weighed and dried at room temperature for a period not less than two (2) hours in a vacuum at a pressure of not more than two hundred (200) microns. The sample shall be reweighed and the loss shall be calculated. Any sample failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.2.2 Weight. The cathode shall be weighed in an analytical balance. Any cathode failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.2.3 Composition. A sample shall be checked by an approved quantitative analysis method for the correct percentage of materials. Any sample failing to yield the proper percentage of material shall be classed defective (Destructive test).

4.5.3 Bulkhead to case force test. The assembly shall be examined for the specified requirement by monitoring the specified depth at the required force. Any assembly that fails to achieve either the force or depth requirements shall be classed defective (Non-destructive test).

4.5.4 Support ampule hardness test. The coupon shall be tested for the specified requirement using a durometer. The coupon shall be classed defective if it fails to comply with all of the specified requirements (Destructive test).

4.5.5 Cup and anode assembly.

4.5.5.1 Weld strength. The cup and anode assembly shall be pulled apart to separation. If the parts separate without indicating a tear or a definite nugget the welded assembly shall be classed defective (Destructive test).

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4.5.5.2 Vacuum pull test. The assembly shall be secured and a vacuum probe shall be applied to the welded surfaces for a minimum period of five tenths (0.5) seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective (Non-destructive test).

4.5.6 Cover and collar assembly.

4.5.6.1 Weld strength. The cover and collar assembly shall be pulled apart to separation. if the parts separate without indicating a tear or a definite nugget, the welded assembly shall be classed defective (Destructive test).

4.5.6.2 Vacuum pull test. The assembly shall be secured and a vacuum probe shall be applied to the welded surfaces for a minimum period of five tenths (0.5) seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective. (Non-destructive).

4.5.7 Separator salt content. The separator salt content shall be checked by quantitative analysis of the disc. The analysis shall include a positive test for a presence of potassium as a major caution. The disc shall be classed defective if it fails to comply with all of the specified requirements (Destructive test).

4.5.8 Presense of ammonia in ampule. The ampule, prior to insertion into the case shall be visually monitored or sensed by an approved system of inspection equipment. Any ampule that is cracked, low in ammonia, contains discolored ammonia, or empty shall be classed defective (Non-destructive test).

4.5.9 Battery, primary, reserve.

4.5.9.1 Case seal and pre-initiation voltage. The sealed battery shall be placed in a helium atmosphere at a pressure of 60 psi for a minimum period of two(2) hours. After removal from the helium atmosphere, battery leak rate shall be determined using a helium mass spectrometer leak detector. Any battery failing to meet the leak rate requirements specified in 3.5.1 shall be rejected.

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4.5.9.2 Activation. The battery shall be pre-conditioned at the applicable test temperature, placed in an approved test fixture and rotated about the Y-Y' axis (See Fig. 2) at thirty six hundred plus or minus two hundred (3600 ± 200) revolutions per minute.

Surface "A" (See Fig. 2) of the battery shall be deformed while rotating at the specified velocity using an approved activation system.

Deformation of surface "A" shall be within twenty-five hundredths (.025") inches of the X-X' reference. (See Fig 2). The battery shall be deformed forty to fifty-two hundredths inch (.040" to .052") by a spherical surface, or ball of a seven-thirty seconds plus or minum on-thirty second inch (7/32 ± 1/32) inch diameter.

The rotation shall be completely stopped at two plus or minus one (2.0 ± 1.0) seconds after deforming surface "A".

A battery that fails to activate on the spin fixture shall be removed and examined for a "No-test" condition. A replacement battery may be activated immediately. Test results for the replacement battery shall be used for acceptance only when an analysis determines the failure to be related to the activating equipment. A "No-test" battery must have its filled ampule intact and all its components present and properly assembled and cause of non-activation must be attributed to the approved test equipment.

Transfer time from the preconditioning temperature chamber to the test fixture shall be such that the test sample temperature shall not vary more than five degrees Centrigrade (5°C). Any battery failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.9.3 Rise time. The battery output voltage shall be continuously monitored for the total test time. The test circuit shown in Figure 3 for the effective eight hundred (800) ohm load shall be used as the test load. The battery shall be classed defective if it fails to comply with all of the specified requirements. The sample shall be activated in accordance with the method specified in 4.5.9.2 (Destructive test).

4.5.9.4 Short time capacity. The battery shall be activated in accordance with 4.5.9.2, and measured for its rise time in accordance with 4.5.9.3. The battery shall be monitored a minimum of once every fifteen (15) minutes for the output requirements on an approved data acquisition system with the test circuit shown in Figure 3 being used for the effective eight hundred (800) ohm test load for a minimum of four (4) hours. The battery shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test)

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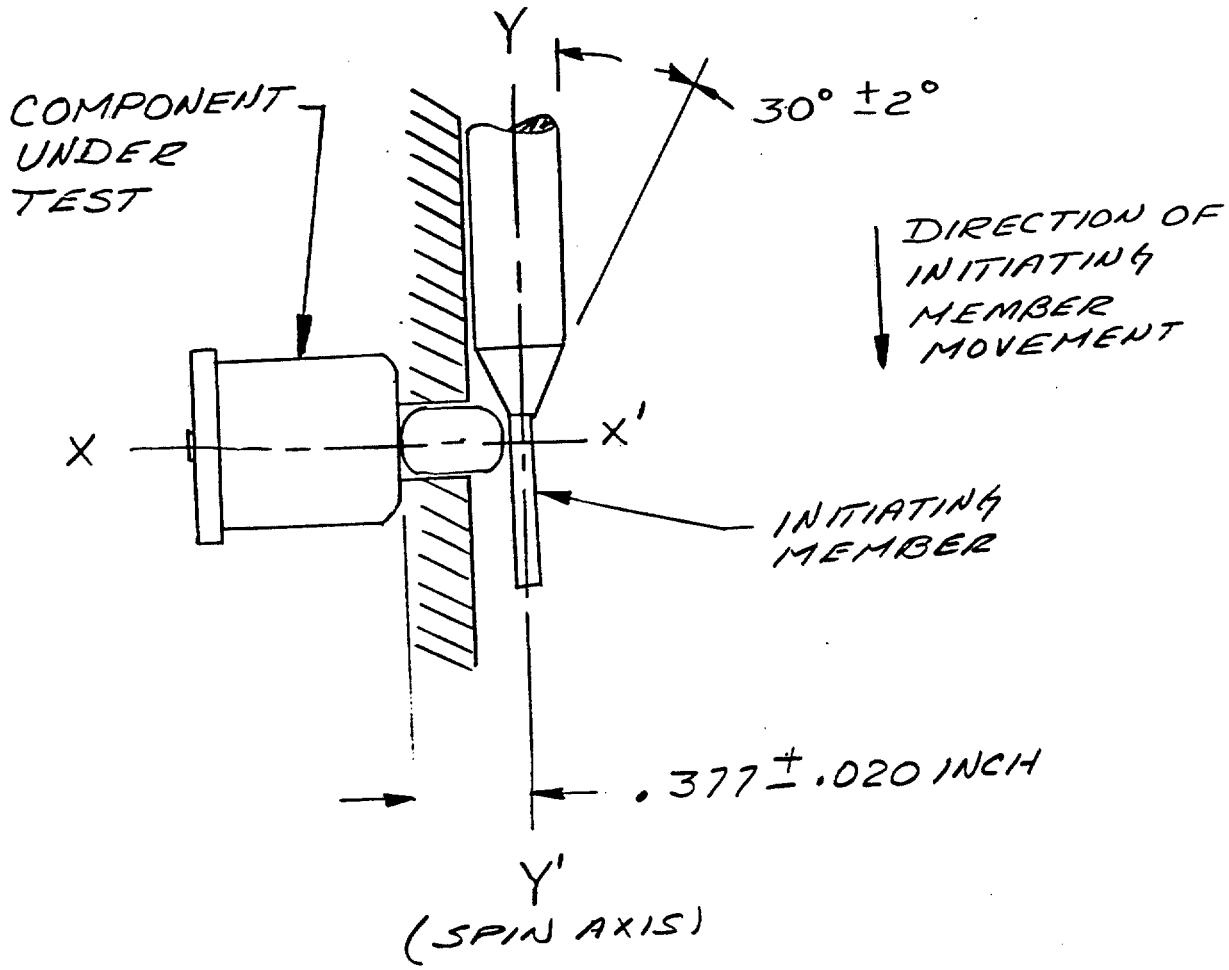


FIGURE 2 : BATTERY ACTIVATION FIXTURE

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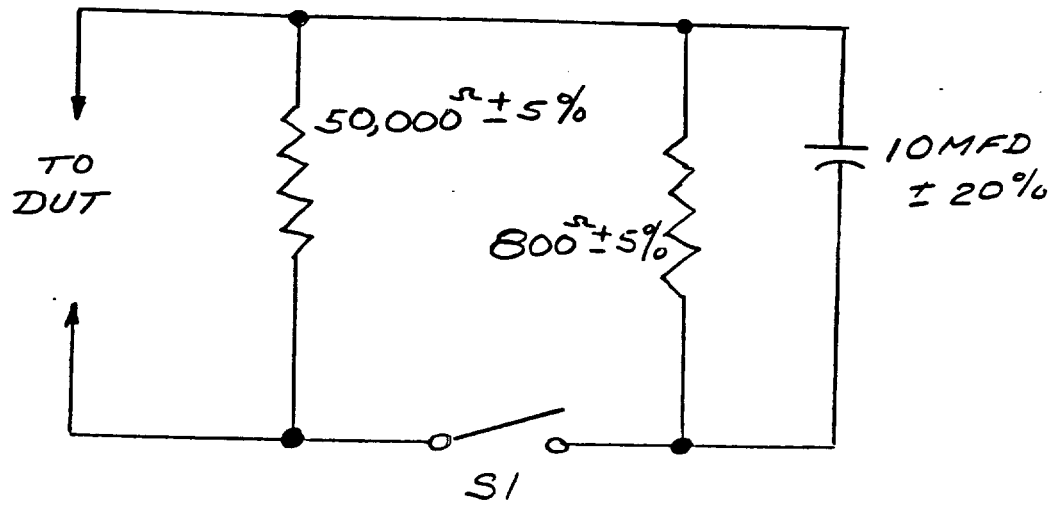


FIGURE 3 : BATTERY DISCHARGE CIRCUIT

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4.5.9.5 Long term capacity. The battery shall be activated in accordance with 4.5.9.2 and measured for its rise time in accordance with 4.5.9.3 at the required test temperature.

The battery shall be loaded with the effective test load shown in fig. 3 for four hundred and five seconds plus twenty (405 + 20) seconds after activation at the required test temperature. The test load shall then be switched to the fifty thousand (50,000) ohm load shown in fig. 3.

The output voltage shall be monitored a minimum of once an hour on a data acquisition system for a minimum of forty eight (48) hours.

The sample shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test).

During first article acceptance testing, the contractor shall monitor the battery voltage a minimum of once an hour until it drops below two and seven tenths (2.7) volts.

4.5.10 Environmental and mechanical tests.

4.5.10.1 Thermal shock. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 113 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirements (Non-destructive test).

4.5.10.2 Temperature cycling. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 105 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirements. Note the humidity requirements are not required for testing.

4.5.10.3 Transportation vibration. The battery shall be tested in accordance with Procedure I, Test Method 119 of MIL-STD-331 (except test vehicle shall be randomly located in standard shipping pack). The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirement (Non-destructive test). Test temperature groups shall be minus fifty two centegrade plus or minus two degrees centigrade ($-52^{o}C \pm 2^{o}C$) thirty degrees centigrade plus ten degrees centigrade ($30^{o}C + 10^{o}C$) and sixty degrees centigrade plus or minus two degrees centigrade ($+60^{o}C \pm 2^{o}C$).

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4.5.10.4 Drop test. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 111 of MIL-STD-331 except the height of drop shall be seven (7) feet. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective it is fails to comply with all of the specified requirements (Non-destructive test).

4.5.10.5 Mechanical shock. The battery shall be mounted in an approved test vehicle and subjected to a seven (7) gun firing utilizing an approved soft catch method. The battery shall not be activated during test (Non-destructive test).

4.5.11 Repair. Any battery undergoing repair shall have the procedures submitted to the technical agency listed in 6.5 prior to performing the necessary work. Any battery repaired without prior Government approval shall be considered as non-conforming material and classed defective (Non-destructive procedure).

4.5.12 Process controls. All essential process control procedures shall be documented and submitted to the technical agency listed in 6.5. Any battery component, sub-assembly or assembly undergoing a process control deemed necessary to assure a reliable end item by the Government that has not been previously submitted to the Government shall be classed as non-conforming material and classed defective. All changes to these process controls must be submitted to the Government prior to the incorporation of the process control.

4.5.13 Dimensional control of molded or plastic parts. In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in 4.4.2.1 through 4.4.2.14 after continuous production of each 5000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he

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presents objective evidence to the contracting officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to the Government for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc mins) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to the Government for approval. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.14 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, the portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. The batteries shall be preserved and packaged in accordance with MIL-STD-1169.

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5.2 Packing.

5.2.1 Level C. The batteries preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. Marking shall comply with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The components and assembly covered by this specification are intended for use as the power supply for the housing, timing and fuzing for the projectile, 155MM, M692/M731.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in section 2.1.

6.3 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report detail in accordance with Data Item Description DI-R-1724 (tailored) giving all detailed inspection and test results for test performed as required by 4.4.3.7. Test report format shall be submitted to the technical agency listed in 6.5.

6.4 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.5 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-I (D), Picatinny Arsenal, New Jersey 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and tests paragraph number. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Subject term (keyword) listing.

Fuze
Moisture
Anode
Cathode

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing activity:
Army-AR

(Project: 1345-A617

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-48343A (AR)	930311

3. DOCUMENT TITLE: BATTERY, PRIMARY RESERVE, AMMONIA, 5 MILLIAMPERE HOUR CAPACITY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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